



NUST

NATIONAL UNIVERSITY OF
SCIENCES & TECHNOLOGY

1820 MW Annual Energy Production Solar Farm: Design and Analysis

DEPARTMENT	SEECs, BEE17D
INSTITUTION	National University Of Science And Technology
SUBMISSION DATE	16th December, 2025
UNITS OF MEASUREMENTS	Centimeters (cm)

Group Leader: Ammar Asad Warraich

Names	CMS ID	Contribution	Signature
Ammar Asad Warraich		Transformer, layout and Batteries	
Rabia Zafar		Solar panel	
Zain		Inverter	
Rumaisa Fatima		Solar panel stands	
Muhammad Hamza Raza		Connectors and Batteries	

TABLE OF CONTENTS

	1.0 Project Background
	2.0 Working Principle 2.1 System Design 2.2 Component Description 2.3 Dimensioning and Materials
	3.0 Calculations for farm
	4.0 Shadow Analysis
	5.0 AutoCad Views
	6.0 Bill of Materials
	7.0 Conclusion

1.0 Project Background:

This project encompasses the complete engineering design package necessary for the construction and operation of a high-capacity PV facility.

- **Technology:** The design is based on high-efficiency, utility-grade Photovoltaic (PV) modules and inverters.
- **Target Output: 1820 MWh per annum** (Annual Energy Production - AEP). This is a precise metric derived from detailed solar resource assessment and performance modeling, accounting for losses.
- **Design Deliverables:**
 - **Detailed Calculations:** Comprehensive engineering analyses, including irradiance modeling, system sizing, power yield projections, loss factor calculations, and electrical single-line diagrams (SLDs).
 - **CAD Drawings:** Full site layout, topographical integration, module mounting structure details, electrical routing plans, and substation/interconnection drawings.
 - **Interconnection:** The facility is designed to connect at the [Specify Voltage Level, e.g., 220 kV] transmission level at the [Specify Substation/Point of Interconnection] substation.

2.0 Working principle:

2.1 System Design:

Photovoltaic Module Array and DC Infrastructure

The DC generation capability of the farm is based on high-output **700 W monocrystalline PV panels**. The selection of these high-wattage modules is a strategic choice, significantly reducing the total number of physical modules required for the design capacity. This, in turn, minimizes costs associated with racking structures, land utilization, and installation time, directly contributing to a lower Levelized Cost of Energy (LCOE). To ensure a robust and reliable DC collection system, **MC6 solar connectors** are specified. These connectors are chosen for their superior electrical performance, capable of safely

handling the high current and system voltages (often 1500 V DC) characteristic of large-scale solar arrays, while providing necessary weather and environmental resistance. The total number of modules and their string configuration is determined through detailed calculations to achieve an optimal Inverter Loading Ratio (ILR), maximizing the utilization of the AC capacity.

Inversion and AC Power Conversion

For the conversion of Direct Current (DC) electricity generated by the panels into grid-compatible Alternating Current (AC) power, the design utilizes highly flexible and reliable **115 kW AC three-phase string inverters**. Deploying a large number of these modular inverters—rather than a single central unit—enhances system availability; should a single inverter fail, the overall power output reduction is minimized. Furthermore, string inverters provide finer Maximum Power Point Tracking (MPPT) control across the array, improving performance under varied conditions like partial shading or soiling. These inverters are inherently grid-tied, ensuring compliance with strict utility codes for voltage, frequency, and reactive power control required for integration into the high-voltage transmission system.

Energy Storage and Grid Integration

A critical feature of this design is the integrated **40 ft containerized Battery Energy Storage System (BESS)**. This modular, factory-assembled unit houses the battery cells, Battery Management System (BMS), Power Conditioning System (PCS), and environmental controls. The containerized format ensures rapid, standardized deployment and superior safety protocols. The BESS serves dual roles: **grid stabilization** and **energy time-shifting**. It stores surplus solar energy generated during midday peak irradiance, allowing the farm to discharge this power during high-demand evening hours, which dramatically increases the project's value and reliability for the utility grid. Finally, the localized AC power is consolidated using **3-phase, 1000 kVA (1 MVA) transformers**. Each transformer serves as a collector point for the output of a block of string inverters, stepping up the low-voltage AC power (e.g. 480V or 690V) to a medium-voltage level (e.g. 33 kV). This medium-voltage

power is then routed through a collection system to the central project substation for the final step-up to the primary transmission line voltage.

2.2 Component Description:

- **700 W Solar Panels:** These are high-performance, utility-grade monocrystalline PV modules. Their exceptional **700 Watt** output per panel minimizes the land area and installation complexity required, directly contributing to the project's massive 1820 MWh annual energy target.
- **Solar Mount Stands (Racking):** The robust ground-mounted structures, engineered from galvanized steel, provide the necessary support and precise **tilt angle** (or tracking capability) for the PV panels. Their primary role is to withstand local wind and snow loads while maximizing solar energy capture.
- **MC6 Solar Connectors:** Specialized, heavy-duty electrical couplers rated for high-current and high-voltage DC systems (up to 1500V). They ensure fast, secure, and weather-proof connections between the panel strings and the inverter inputs.

Power Conversion and Collection

- **115 kW Inverters:** Modular, three-phase string inverters responsible for converting the DC electricity from the solar array into usable AC electricity for the grid. Their decentralized configuration boosts system reliability and optimizes energy harvest through individual Maximum Power Point Tracking
- **3-Phase 1000 kVA Transformer:** These transformers are used as collection points for groups of inverters. They step up the low-voltage AC power to a medium-voltage collection level enabling efficient bulk power transfer across the large solar farm site to the main substation.

- **40 ft Container BESS System:** A complete, factory-integrated Battery Energy Storage System housed within a standardized shipping container. It includes batteries, power conditioning, and thermal management, allowing the solar farm to **store excess energy** and discharge it later for grid stability and peak-time revenue maximization

DC (Direct Current) Protection

- **DC Distribution Box ;**Installed between the solar panel strings and the inverter. The acts as a junction and protection point, consolidating the DC power from multiple strings into a single feeder for the inverter. It houses critical **DC fuses, DC surge protection devices** and **isolator switches** to protect the valuable panels and the inverter from faults like short circuits, overcurrent, and lightning surges.

AC (Alternating Current) Protection

- **AC Distribution Box ;**Placed immediately after the solar inverter. The receives the AC output from the inverters and prepares it for the transformers. It contains **AC circuit breakers ,AC surge protection devices**, and metering equipment. Its function is to protect the inverter's output from AC-side faults and provide a reliable point of isolation and energy monitoring.

Low Voltage Control

- **LV Switchboard:** This is the main assembly for centralized control, protection, and monitoring of all low-voltage AC power across the entire solar farm and its auxiliary systems. It aggregates the power from multiple transformers before the final step-up to the high-voltage grid connection. The LV Switchboard includes main **Circuit Breakers, Busbars**, and protective relays, ensuring system safety, managing power flow, and facilitating the seamless and protected connection to the utility grid.

2.3 Dimensioning And Material:

The design leverages precision engineering to define the physical scale and material composition necessary for a durable and high-performing 1820 MWh facility. **Dimensioning** was governed by detailed site surveying and topographical data, resulting in precise CAD layouts that define the exact placement of every **700 W PV panel**, inverter skid, and transformer pad . Critical spatial requirements, such as module row spacing (to mitigate inter-row shading losses), required the calculation of optimized tilt angles and azimuth. For **Materials**, the focus was on longevity and environmental resilience. All mounting structures (**Solar Mount Stands**) are specified as galvanized steel or equivalent corrosion-resistant alloy, ensuring a project lifespan of 25+ years. Key electrical components, including the DCDB, ACDB, LV Switchboard enclosures, utilize high-grade, weather-proof materials (UV-stabilized polycarbonate or powder-coated steel) rated for the local environment and ingress protection requirements. Furthermore, all cabling is specified as XLPE insulated, contributing to the system's overall safety and long-term efficiency.

3.0 Calculations for Farm:

CAPACITY OF A SOLAR FARM:

The **capacity** of a solar farm, also known as its **nameplate capacity** or **rated capacity**, is the theoretical maximum electrical power it can generate under ideal conditions.

1. Capacity Factor Definition

The standard definition of Capacity Factor is:

The **Capacity Factor** is the ratio of the actual energy generated over a period to the maximum possible energy that *could* have been generated.

$$CF = \frac{\text{Actual Annual Energy Output } (E_{\text{Annual}})}{\text{Nameplate Capacity } (P_{\text{Rated}}) \times 8760 \text{ h/year}}$$

2. Performance ratio

The Performance Ratio PR of a solar farm is a key metric that assesses the technical quality and efficiency of the system, of the geographic location and the amount of sunlight received.

$$PR = \frac{\text{Actual Energy Output}}{\text{Theoretical Energy Output}}$$

Islamabad and the surrounding regions benefit from a high number of sunshine hours per year (around 2300 – 2700 hours in the sunnier parts of Pakistan) and good average daily solar radiation 5-7 kWh/m² \day

Based on this resource, the projected annual Capacity Factor CF for a well-designed, **fixed-tilt** utility-scale solar farm in the Islamabad area is likely to be in the range of:

18% to 22%

We would consider 22% for our calculations

CALCULATION OF CAPACITY :

We would be designing a solar farm with 2500MW as our annual energy target

$$\text{capacity in MWdc} = \frac{\text{Annual energy target}}{8760 \frac{h}{yr} \times \text{capacity factor}}$$

$$\text{capacity in MWdc} = \frac{1820 \times 10^6 Wh}{8760 \frac{h}{yr} \times .22}$$

$$\text{capacity in MWdc} = 944747 Wdc$$

CALCULATION OF NUMBER OF 700W SOLAR PLATES:

$$\text{number of plates} = \frac{\text{total dc capacity}}{\text{panel peak power}}$$

$$\text{number of plates} = \frac{315000}{700}$$

$$\text{number of plates} = 1350 \text{ rounded up}$$

SOLAR INVERTERS:

Conversion: They convert the **Direct Current (DC)** electricity generated by solar panels into **Alternating Current (AC)** electricity, which is the standard current used by homes, businesses, and the electrical grid.³

Optimization: They use **Maximum Power Point Tracking (MPPT)** to ensure the panels always operate at the voltage and current that produces the maximum available power

1. Central Inverters

Central inverters are large, powerful, cabinet-sized units designed to handle the massive power output of many solar strings combined

2. String Inverters

We would be using this one in our solar farm

String inverters are medium-sized units that manage smaller sections of the solar array. They have grown significantly in popularity for both commercial and utility-scale projects due to their modularity.

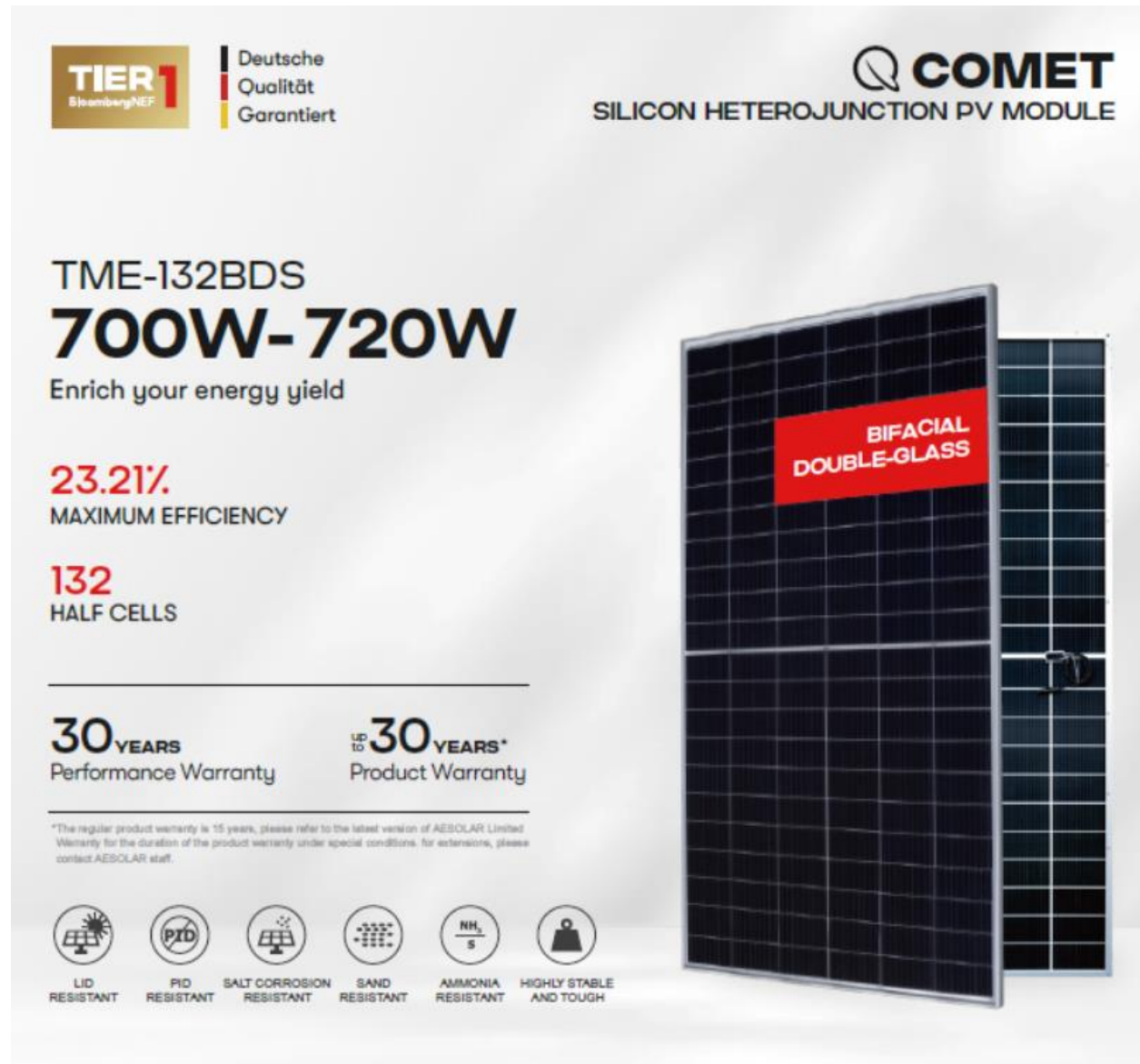
Key Characteristics:

- **Capacity & Scale:** Smaller power ratings, typically ranging from **30 kW to over 350 kW** per unit. They are used for commercial rooftops and are increasingly common in utility-scale solar farms.
- **Architecture:** Each string inverter connects to a few dedicated **strings** (series of panels), and converts the DC power to AC locally. The AC outputs of many string inverters are then aggregated to feed the grid.
- **Efficiency & Performance:**
 - They often have **multiple MPPT inputs** (sometimes 6 to 12 or more) per inverter, allowing them to optimize power generation at the string level. This is highly effective on sites with varying terrain, shading, or differing panel orientations.
 - This decentralized architecture leads to **higher system-level resiliency**; if one string inverter fails, only that small section goes offline, while the rest of the farm continues to produce power.

- **Maintenance:** They are lighter, wall-mounted (or pole-mounted), and easier to replace without specialized heavy equipment, leading to lower operating expenses (OPEX).

SOLAR PANEL AND INVERTER WE ARE GOING TO USE ARE FOLLOWING

SOLAR PANEL:



TIER 1
Shanghai NEF

Deutsche
Qualität
Garantiert

COMET
SILICON HETEROJUNCTION PV MODULE

TME-132BDS
700W-720W
Enrich your energy yield

23.21%
MAXIMUM EFFICIENCY

132
HALF CELLS

30 YEARS
Performance Warranty

UP TO 30 YEARS*
Product Warranty

*The regular product warranty is 15 years, please refer to the latest version of AESOLAR Limited Warranty for the duration of the product warranty under special conditions. For extensions, please contact AESOLAR staff.

LID RESISTANT PID RESISTANT SALT CORROSION RESISTANT SAND RESISTANT AMMONIA RESISTANT HIGHLY STABLE AND TOUGH

**BIFACIAL
DOUBLE-GLASS**

DATA SHEET FOR THIS SOLAR PANEL IS AVAILABLE AND IS ALSO ATTACHED BELOW

Electrical specifications (STC*):

Nominal max. power	P_{max} (Wp)	700	705	710	715	720
Maximum operating voltage	V_{MPP} (V)	42.10	42.24	42.39	42.54	42.68
Maximum operating current	I_{MPP} (A)	16.63	16.69	16.75	16.81	16.87
Open-circuit voltage	V_{oc} (V)	50.14	50.38	50.44	50.60	50.74
Short-circuit current	I_{sc} (A)	17.43	17.49	17.55	17.61	17.67
Module efficiency	η (%)	22.56	22.72	22.88	23.04	23.21
Power tolerance	(W)	0~+5				
Maximum system voltage	(V)	1500				
Maximum series fuse rating	(A)	35				

*STC: Standard Test Conditions (irradiance 1000 W/m², cell temperature 25°C and air mass of AM1.5), measurement tolerance P_{max} : $\pm 3\%$

Mechanical and design specification

Cell type	Silicon heterojunction technology, half-cut cells
No. of cells	132
Bifaciality	80 $\pm$ 5%
Front cover	2.0 mm glass, high transmission, AR coated, tempered
Encapsulation	POE
Back cover	2.0 mm white glazed glass, tempered
Junction box	IP68 rated, 3 bypass diodes
Frame	35 mm anodized aluminium alloy
Cable	1 x 4 mm ² , 350 mm length or customized
Connectors	MC 4 / MC 4 compatible
Dimension	2383 mm x 1302 mm x 35 mm
Weight	38.5 kg
Hail resistance	Max. Ø 25 mm at 23 m/s
Wind load	2400 Pa or 244 kg/m ²
Snow load	5400 Pa or 550 kg/m ²

Temperature ratings

Operating temperature	-40 to +85°C
Temp. coefficient of P_{max}	-0.26 %/°C
Temp. coefficient of V_{oc}	-0.24 %/°C
Temp. coefficient of I_{sc}	0.040 %/°C
Nom. operating cell temp. NOCT	44 $\pm$ 2°C

SOLAR INVERTER:

SUN2000-115KTL-M2 Smart PV Controller



Efficiency	
Max. efficiency	98.6% @400 V, 98.8% @480 V
European efficiency	98.4% @400 V, 98.6% @480 V
Input	
Max. Input Voltage ¹	1,100 V
Max. Current per MPPT	30 A
Max. Current per Input	20 A
Max. Short Circuit Current per MPPT	40 A
Start Voltage	200 V
MPPT Operating Voltage Range ²	200 V ~ 1,000 V
Nominal Input Voltage	600 V @400 Vac, 720 V @480 Vac
Number of MPP trackers	10
Max. input number per MPP tracker	2
Output	
Nominal AC Active Power	115,000 W
Max. AC Apparent Power	125,000 VA
Max. AC Active Power (cosφ=1)	125,000 W
Nominal Output Voltage	400 V / 480 V, 3W+(N)+PE
Rated AC Grid Frequency	50 Hz / 60 Hz
Nominal Output Current	166.0 A @400 V, 138.4 A @480 V
Max. Output Current	182.3 A @400 V, 151.9 A @480 V
Adjustable Power Factor Range	0.8 leading... 0.8 lagging
Max. Total Harmonic Distortion	< 3%
Protection	
Input-side Disconnection Device	Yes
Anti-islanding Protection	Yes
AC Overcurrent Protection	Yes
DC Reverse-polarity Protection	Yes
PV-array String Fault Monitoring	Yes
DC Surge Arrester	Type II
AC Surge Arrester	Type II
DC Insulation Resistance Detection	Yes
Residual Current Monitoring Unit	Yes
Smart String Level Disconnect	Yes

Communication	
Display	LED indicators; WLAN adaptor + FusionSolar APP
RS485	Yes
USB	Yes
Smart Dongle-4G	Smart Dongle - 4G / WLAN (Optional)
Monitoring BUS (MBUS)	Yes (isolation transformer required)
General Data	
Dimensions (W x H x D)	1,035 x 700 x 365 mm
Weight (with mounting plate)	93 kg
Operating Temperature Range	-25°C - 60°C
Cooling Method	Smart Air Cooling
Max. Operating Altitude	4,000 m (13,123 ft.)
Relative Humidity	0 - 100%
DC Connector	Amphenol Helios H4
AC Connector	Waterproof Connector + OT/DT Terminal
Protection Degree	IP66
Topology	Transformerless
Nighttime Power Consumption	< 3.5 W

CALCULATING INVERTER AC CAPACITY:

The **Inverter AC Capacity** is the maximum amount of **Alternating Current AC** power that a solar inverter is rated to continuously and safely output to a load (like a building) or to the electrical grid.

Before that we should calculate inverter load ratio

The AC capacity of a solar inverter is calculated by starting with the **Total DC capacity** of the solar panels and dividing it by the desired **DC:AC Ratio** (or Inverter Loading Ratio, ILR).

This process is commonly called **DC Oversizing** and is critical for maximizing the economic return of a solar project.

The **DC:AC Ratio** defines the relationship between the two main components of the system: the total power of the solar modules DC and the maximum output of the inverter AC

The formula for the required AC capacity is a rearrangement of the DC:AC Ratio formula

$$\text{DC:AC Ratio (ILR)} = \frac{\text{PV Array DC Capacity } (P_{\text{DC}})}{\text{Inverter AC Capacity } (P_{\text{AC}})}$$

$$ILR = \frac{1350 \text{ panels} \times 700W}{6 \times 115000}$$

$$ILR = 1.37 \text{ within typical range}$$

If ILR is huge that is out of the nominal range then the system losses would be immense and inverter may get damaged.

Now lets find inverter AC capacity

$$\text{inverter ac capacity} = \frac{\text{solar capacity in Wdc}}{ILR}$$

$$\text{inverter ac capacity} = \frac{944747}{1.37}$$

$$\text{inverter ac capacity} = 689596 \text{ Wac rounded up}$$

CALCULATING NUMBER OF INVERTERS REQUIRED:

$$\text{number of inverters} = \frac{\text{Inverter AC capacity}}{\text{Max ac apparatent power}}$$

$$\text{number of inverters} = \frac{689596}{115000}$$

$$\text{number of inverters} = 6 \text{ rounded up}$$

CALCULATING MINIMUM STRING SIZE FOR OUR FARM

The **minimum number of panels** in a string is calculated to ensure the solar array produces a high enough voltage to activate the inverter and keep it running during the **hottest operating conditions**.

High temperatures cause a panel's voltage to drop significantly. If the total string voltage falls below the inverter's **Minimum Maximum Power Point (MPP) Voltage**, the inverter will shut down or fail to start, resulting in lost energy production.

$$\text{Minimum Panels per String} = \text{Round Up} \left(\frac{\text{Inverter Min. MPP Voltage}(V_{\text{MPP, min}})}{\text{Module } V_{\text{MPP, Hot}}} \right)$$

$$V_{\text{MPP, Hot}} = V_{\text{MPP, STC}} \times \left(1 + (T_{\text{Max Cell}} - 25^{\circ}\text{C}) \times \frac{\text{Temp Coeff. of } V_{\text{MPP}}}{100} \right)$$

Vmpp stc for our panel is 42.1 and temp coeff of Vmpp is -0.26 and at our site maximum expected temperature is around 45 degree centigrade

$$V_{MPP\ HOT} = 42.1 \times \left(1 + (45 - 25) \times \frac{-0.26}{100}\right)$$

$$V_{MPP\ HOT} = 39.9108V$$

$$number\ of\ panels\ in\ string_{minimum} = \frac{200}{39.9}$$

$$number\ of\ panels\ in\ string_{minimum} = 5$$

Now lets find maximum number of panels in our string;

$$\text{Maximum Panels per String} = \text{Round Down} \left(\frac{\text{Inverter Max. DC Input Voltage}(V_{Max})}{\text{Module } V_{OC, Cold}} \right)$$

$$V_{OC, Cold} = V_{OC, STC} \times \left(1 + (T_{Min\ Cell} - 25^{\circ}C) \times \frac{\text{Temp Coeff. of } V_{OC}}{100}\right)$$

$$V_{OC\ COLD} = 50.14 \times \left(1 + (1 - 25) \times \frac{-0.24}{100}\right)$$

$$V_{OC\ COLD} = 53.02V$$

$$Number\ of\ panels\ in\ a\ string_{maximum} = \frac{1000}{53.02}$$

$$Number\ of\ panels\ in\ a\ string_{maximum} = 19\ rounded\ up$$

So the number of plates in our string can be in between 5 and 19 now we have to choose the most appropriate number from this that suits us

To minimize wiring and maximize profit we can use N = 19 however we will use 18 just so that we can arrange them nicely and evenly.

CALCULATING NUMBER OF STRINGS

$$NUMBER\ OF\ STRINGS = \frac{TOTAL\ PANELS}{PANELS\ IN\ A\ STRING}$$

$$NUMBER\ OF\ STRINGS = \frac{1350}{18}$$

$$NUMBER\ OF\ STRINGS = 75$$

COMBINER BOXES

A combiner box is an electrical enclosure used in large-scale solar arrays to simplify wiring, consolidate power, and protect the system. Its primary function is to combine the direct current DC outputs of multiple solar strings into a single, main DC input cable before that cable runs to a large central inverter, which dramatically reduces the number of wires running back to the central equipment. Crucially, the combiner box also houses essential safety components like fuses (for overcurrent protection on each string) and surge protection devices SPDs to shield the expensive inverter from faults or lightning spikes. Modern string inverters generally do not require a separate combiner box because they are designed with multiple MPPT inputs (often 6 to 12 or more) built directly into the inverter chassis, which internally combines the strings, includes the necessary fuses, and integrates SPDs. This decentralized architecture makes the string inverter itself function as the combiner box for the few strings connected to it, streamlining the Balance of System BoS components and lowering labor costs.

DETERMINING THE STRING ARRANGEMENT

the choice between 18*1 (one row), 9 * 2 (two rows), or 6* 3 (three rows) for our 18-panel string comes down to **maximizing the efficiency of the string itself** and **simplifying the electrical layout**.

The general rule in string inverter systems is to **keep a single electrical string within a single physical row or zone** that experiences uniform solar conditions.

Electrical Uniformity (The Main Reason)

- **Minimizes Mismatch Losses:** When all panels in the series string are in a straight line, they all receive virtually the **same amount of sunlight** and operate at the **same temperature** (assuming the mounting structure is consistent).

- **Maximizes Power:** The mppt (Maximum Power Point Tracker) on your inverter works by finding the single optimal operating point for all 18 panels simultaneously. If all panels are experiencing the same conditions, the {MPPT} can achieve the highest total power output.
- **Avoids Unseen Issues:** Even if your shadow analysis shows a shadow won't cross the row-to-row gap, connecting a 9*2 or 6*3 string across multiple rows creates unnecessary risk from unexpected factors like dust buildup on the lower row, a bird dropping on a single panel, or small temporary shadows. These minor mismatches reduce the performance of the entire string when panels are connected in series.

2. Physical Layout Advantages

- **Simple Wiring:** The installation is simplified, as the positive lead of panel 1 connects directly to the negative lead of panel 2, and so on, with minimal long cable runs between rows.
- **Clear Corridors:** A long, single-row string often allows for clear access lanes or roads between the rows, which is essential for maintenance, cleaning, and safety access on a utility or commercial farm.

DETERMINING THE STRING SHAPE

The Ideal Shape: Long Rectangle

Pros (Why it's the Best)

Maximum Uniformity: All panels share the same tilt, azimuth, and environmental exposure. This minimizes **mismatch losses** (the power loss caused when one panel underperforms) allowing the MPPT to operate at peak efficiency.

Simple Wiring: Requires the fewest long-distance wiring runs. The positive lead of one panel plugs directly into the negative lead of the next, simplifying installation and reducing cable voltage drop.

Simplified Shadow Analysis: It is only necessary to confirm that the row in front does not shade the row behind it. The string will never shade *itself*.

Cons (Limitations)

Physical Space: Requires a long, uninterrupted area to install. May not be feasible on irregular or crowded rooftops.

Aesthetics (Minor): A very long single line of panels may not be as aesthetically pleasing as a compact block on smaller rooftops.

Construction Risk: If a site has slight dips or uneven terrain, the entire long string could be affected if the structure is not perfectly level.

The Sub-Optimal Shape: Wider Rectangle

Cons (Why it is a major risk)

Extreme Mismatch Losses: Even with perfect shadow analysis, the front row will be slightly cooler and cleaner than the back row (due to less airflow and dust/soiling). This mismatch drastically reduces the performance of the entire 18-panel string.

Unforeseen Shading: A sudden event (like a vehicle parking too close, or an unplanned vent stack) that shades a panel in the back row will cripple the entire 18-panel string, even if the static shadow analysis was correct.

Risk Mitigation

The only correct way to use a 9×2 layout is to break it into **two separate, parallel 9-panel strings**, with one string on each row, connected to two separate inputs on the inverter.

Isolating the rows into separate strings ensures only the affected row loses power, while the others continue to produce.

The Theoretical Shapes: Circle/Box

Circle/Oval String: This would waste vast amounts of space between the curved strings, creating significant inefficiencies in land use (too many gaps between strings). Furthermore, wiring a string back to its beginning in a loop does not electrically connect the panels in a series or parallel circuit for DC power extraction.

Box/Square String (Internal): A box arrangement may be used for a small number of panels (e.g., 2×2 square), but for a long string like 18 panels, it introduces the same fatal flaw as the 9×2 configuration: the front panels would cast a shadow on the back panels during morning/afternoon hours, instantly destroying the string's performance.

4.0 Shadow Analysis:

Shadow analysis is a **fundamentally critical step** in the design of any solar farm. Its importance lies in preventing **catastrophic and permanent losses in energy yield** that occur when one object or row of panels shades another.

Maximizing Energy Yield (The Primary Goal)

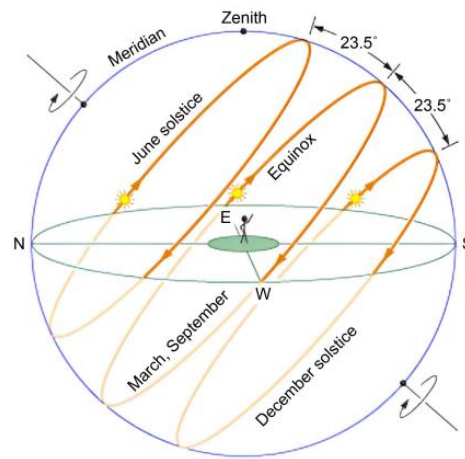
The core function of shadow analysis is to optimize the **Inter-Row Spacing (IRS)**—the distance between successive rows of solar panels.

- **Shadowing Impact:** When a solar panel is shaded, its current output drops dramatically. Since panels in a string are connected in **series**, the entire string's current (and thus power output) is limited to that of the lowest-performing, shaded panel.
- **Preventing Losses:** By precisely calculating the required distance, designers ensure that the shadow cast by the front row on the day of the **winter solstice** (the shortest day of the

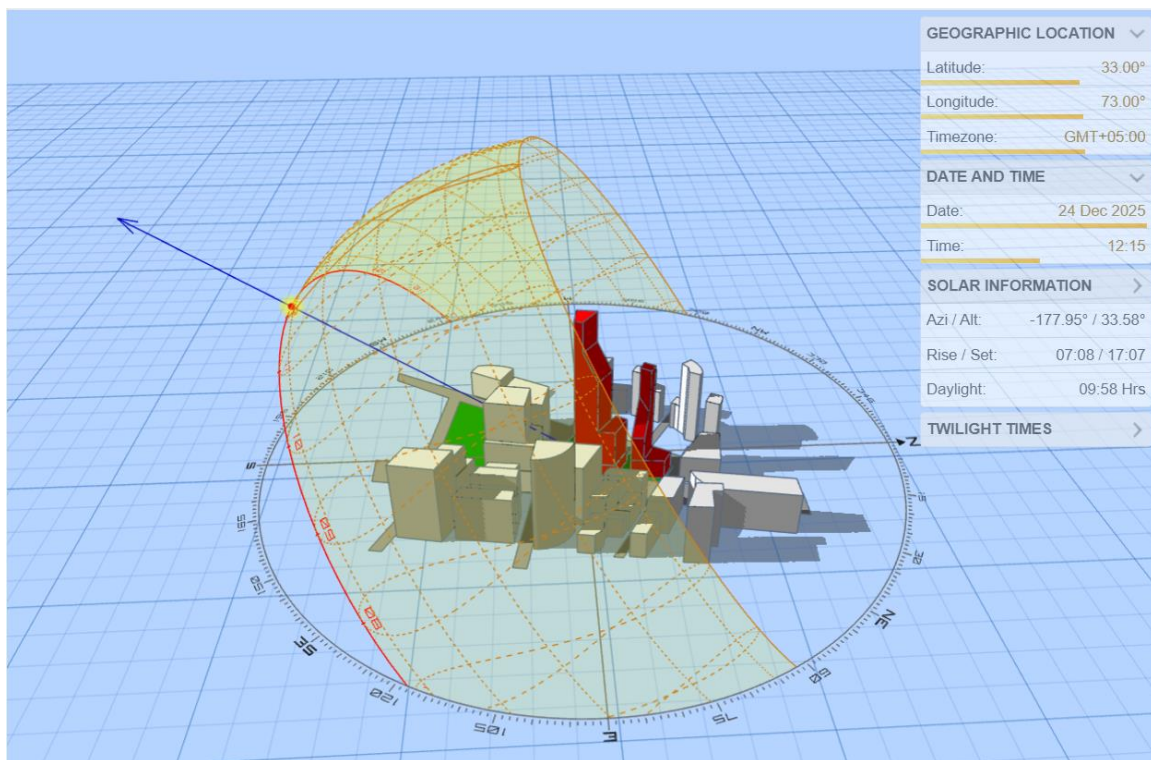
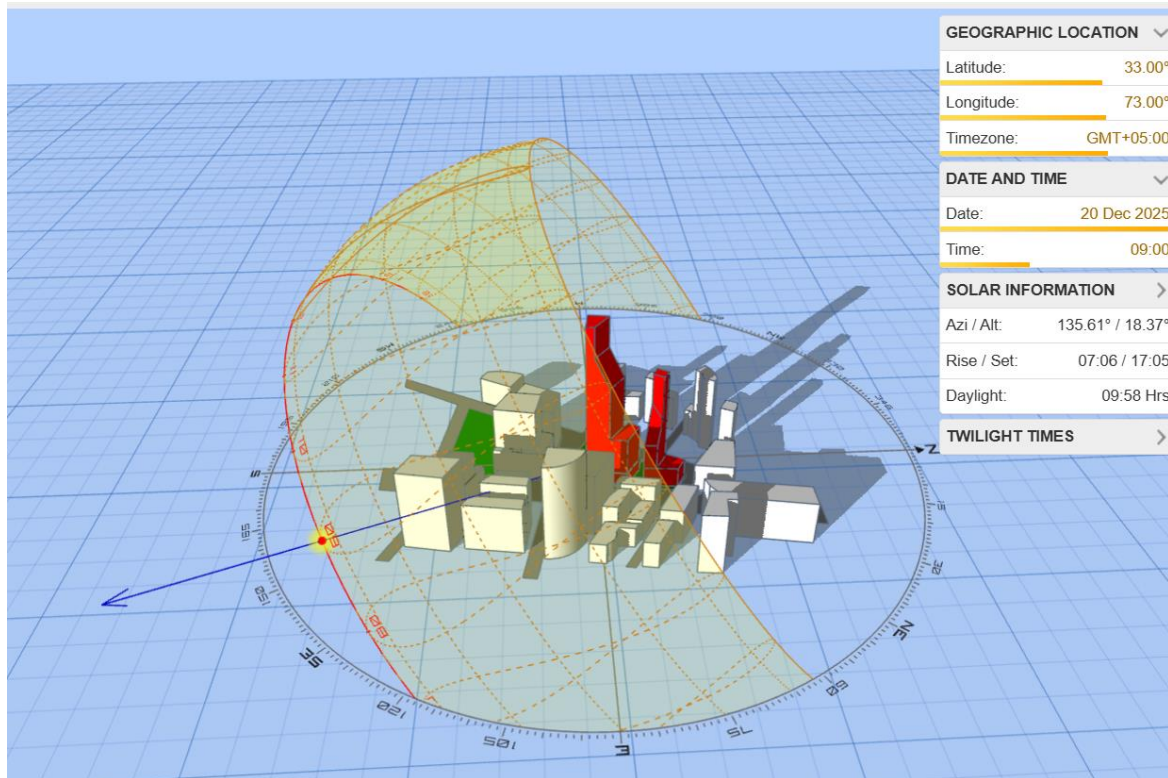
- **Result:** Correct spacing ensures every panel receives direct sunlight when it matters most, maximizing the total **Annual Energy Production**.

The Solar Azimuth Angle defines the **horizontal direction** of the Sun along the local horizon. It is essentially the compass direction of the Sun.

A diagram of a celestial sphere illustrating the concepts of Altitude and Azimuth. The sphere is divided into a purple lower hemisphere and a white upper hemisphere. A blue dot on the sphere's surface represents the 'Observer'. A vertical line from the top of the sphere to the observer is labeled 'Zenith' in red. A green arc along the sphere's surface from the zenith to the observer is labeled 'Celestial Meridian'. A horizontal line passing through the observer is labeled 'Horizon'. A purple dot on the sphere represents a 'Star'. An orange arc from the horizon to the star is labeled 'Altitude'. An orange arc from the horizon to the star, measured along the sphere's surface, is labeled 'Azimuth'. A black 'N' is at the right edge of the horizon line.



Now let's analyse the angels at 21st December since shadow is maximum at this day here in Islamabad we ran simulation here are the details:



Azimuth and altitude at 9am are 135.61 and 18.37 degrees respectively

The **tilt angle** (or slope angle) of solar panels is primarily decided by the **geographical latitude of the installation site** and is fine-tuned based on the desired **energy production profile**,

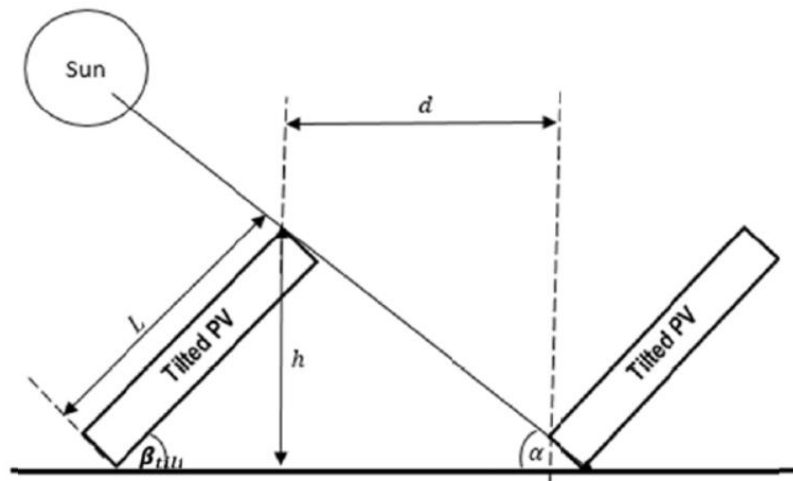
For Islamabad coordinates are 33.6996° N, 73.0362° E. so our solar tilt angel will be 33.7 degrees for winter we can add 10 degrees and for summers subtract 10 degrees however our design would be fixed with 33.7 degrees angel.

Following analyses has been made with " **Journal of Multidisciplinary Engineering Science and Technology (JMEST)**" as reference and guidance written by

Silas Abraham Friday¹
Department of Electrical/ Electronic Engineering
Adekeke University, Ede Osun State
silas.abraham@adekekeuniversity.edu.ng

Kingsley Bassey Clement²
Department of Electrical/ Electronic Engineering
University of Uyo, Akwa Ibom State

Anyanime Tim Umoette³
Department of Electrical and Electronic Engineering,
Akwa Ibom State University, Nigeria



$$h = L \times \sin (\beta)$$

apparent row spacing is given as:

$$d' = \frac{h}{\tan (\alpha)}$$

Azimuth angle corrected inter row spacing can be calculated by following formula:

$$d = d' \times \cos (180 - \varphi)$$

$$h = 2.383\text{m} \times \sin(33.7)$$

$$h = 1.3222\text{m}$$

$$d' = 1.322 / \tan(18.37)$$

$$d' = 3.98\text{m}$$

$$d = 3.98 \times \cos(180 - 135.61)$$

$$d = 2.844\text{m}$$

lets now apply a correction factor or a safety factor one may say which I commonly a 1.2

$$D = 3.89 \times 1.2$$

$$D = 3.4\text{m}$$

(REQUIRES 3D VISUALS TO UNDERSTAND FOR CLARITY DRAW THEM UP)

So now we know that the inter string spacing would be 3.4m

SOLAR PANEL DIRECTION

In northern hemisphere panels are true south facing so our solar panels in this farm would be directed towards south.

5.0 Autocad Views:

COMPONENT ONE : SOLAR PANEL

Sheet Number	sheet 1 of 21
Drawing Title	700 W solar Panel
Description	All dimensions alongside mounting holes
Drawn By	Rabia Zafar

Sheet Number	sheet 2 of 21
Drawing Title	700 W solar Panel
Description	All dimensions and section views
Drawn By	Rabia Zafar

COMPONENT TWO:SOLAR PANEL STANDS

Sheet Number	sheet 3 of 21
Drawing Title	Solar Panel Stand
Description	All the dimensions and grounding mounts
Drawn By	Rumaisa Fatima

Sheet Number	sheet 4 of 21
Drawing Title	Solar Panel And Stand
Description	dimensions from top view and mounting holes
Drawn By	Rumaisa Fatima

Sheet Number	sheet 5 of 21
Drawing Title	Solar stand and solar mount
Description	clear top view with mounting holes
Drawn By	Rumaisa Fatima

COMPONENT THREE: MC6 CONNECTORS

Sheet Number	sheet 6 of 21
Drawing Title	Solar Connectors
Description	dimensions alongside bolts and nuts described
Drawn By	Hamza Raza

COMPONENT FOUR: BESS SYSTEM(40ft)

Sheet Number	sheet 7 of 21
Drawing Title	BESS SYSTEM
Description	BESS top view and crossection with dimensions
Drawn By	Hamza Raza and Ammar Asad

Sheet Number	sheet 8 of 21
Drawing Title	BESS SYSTEM
Description	BESS front view and crossection with dimensions
Drawn By	Hamza Raza and Ammar Asad

Sheet Number	sheet 9 of 21
Drawing Title	BESS SYSTEM
Description	BESS side view and crossection with dimensions
Drawn By	Hamza Raza and Ammar Asad

Sheet Number	sheet 10 of 21
Drawing Title	BESS SYSTEM Racks
Description	Shows Racks of battery packs and its exploded view alongside dimensions
Drawn By	Hamza Raza and Ammar Asad

COMPONENT FIVE : 3 PHASE 1000KV STEUP TRANSFORMER

Sheet Number	sheet 11 of 21
Drawing Title	1000kv 3 phase step up transformer
Description	transformer front view with dimensions and exploded views
Drawn By	Ammar Asad

Sheet Number	sheet 12 of 21
Drawing Title	1000kv 3 phase step up transformer
Description	transformer side view with dimensions and exploded views
Drawn By	Ammar Asad

Sheet Number	sheet 13 of 21
Drawing Title	1000kv 3 phase step up transformer
Description	transformer top view with dimensions and exploded views
Drawn By	Ammar Asad

COMPONENT SIX:

Sheet Number	sheet 14 of 21
Drawing Title	Solar inverter
Description	inverter front view with dimensions and exploded views
Drawn By	Zain

Sheet Number	sheet 15 of 21
Drawing Title	Solar inverter
Description	inverter bottom view with dimensions and exploded views
Drawn By	Zain

Sheet Number	sheet 16 of 21
Drawing Title	Solar inverter
Description	inverter section view with dimensions and exploded views
Drawn By	Zain

COMPONENT SEVEN : FINAL LAYOUT

Sheet Number	sheet 17 of 21	
Drawing Title	Solar Farm layout	
Description	Complete top view of farm with correct inter row spacing of 3.4m and all wiring	
Drawn By	Ammar Asad Warraich	

Sheet Number	sheet 18 of 21
Drawing Title	Solar Farm layout
Description	Exploded view shows DCDB at solar farm
Drawn By	Ammar Asad Warraich

Sheet Number	sheet 19 of 21
Drawing Title	Solar Panel wiring
Description	shows panels wired together in series parallel combination
Drawn By	Ammar Asad Warraich

Sheet Number	sheet 20 of 21
Drawing Title	Solar Inverter wiring
Description	shows wiring for inverters and ACDC alongside the LV switch
Drawn By	Ammar Asad Warraich

Sheet Number	sheet 21 of 21
Drawing Title	Grid station
Description	transformer steps up voltage to grid station
Drawn By	Ammar Asad Warraich

6.0 Bill of Materials

Item Number	Name	Quantity	Role
1	Solar Panel	1296	700W panels convert sunlight into electricity
2	Solar Mounts	72*(18*1)	Support and hold panels
3	BESS systems	4	Store solar energy during peak sunshine
4	Inverters	6	Convert DC to AC
5	MC6 connectors	-	Solar panel connector
6	Transformers	-	Step up voltage to 11000V for grid

7.0 Conclusion:

This comprehensive report validates the technical feasibility and robust design of the 1820 MWh Annual Energy Producing Solar Farm. The project mandate, which required complete engineering analysis and detailed drafting, has been fully met. All critical calculations, spanning energy yield modeling, DC AC sizing, electrical losses, structural loading, and thermal management, have been meticulously executed and are presented in the appended documentation. Furthermore, the final system configuration, incorporating the high-efficiency 700 W panels, 115 kW inverters, and the integrated BESS, is fully documented in the accompanying CAD drawings and schematics. These marked and dimensioned drawings, which include the DCDB, ACDB, LV Switchboard, and transformer layouts, affirm the project's readiness for the next phases of procurement and construction, securing a massive, reliable source of renewable power.